SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

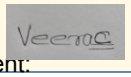
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	SLOT D/2025	**Date:**	10-10-2025

Code of Conduct

I C Veera Bharadwaj Kumar(192565061) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set,	**Q3**

	node expansion	

9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify which statements are correct and which are incorrect. For each incorrect statement, quote the erroneous phrase or claim (one statement contains more than one error).

Statement	Correct / Incorrect	Erroneous phrase / claim
S1		
S2		
S3		
S4		
S5		
S6		

(b) Root cause analysis: for every error, explain the underlying misconception, i.e. which concept has been misunderstood and why the statement goes wrong.

Error (statement / part)	Root cause (misconception)

(c) Rewrite each faulty statement correctly and justify your correction. For a statement that is already correct, write "No change" and justify why.

Statement	Corrected statement	Justification
S1		
S2		
S3		
S4		
S5		
S6		

(d) Apply the corrected concepts to the following case. A self-driving taxi chooses a highway route that every available traffic report showed to be clear; an unforeseeable landslide then blocks the road.

(i) Using the four factors that decide whether an agent is rational, argue whether the taxi's choice was rational.

(ii) Name the properties of the taxi's task environment that make perfect foresight impossible.

(iii) Describe one situation in which a simple reflex taxi would behave poorly, and state what a model-based reflex taxi keeps in order to avoid it.

(i)	
(ii)	

(iii)	

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

A student formulates the **Missionaries and Cannibals** problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bank of a river, with a boat that carries one or two people and cannot cross empty. The boat is on the left bank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). The objective is to move everyone to the right bank using the **fewest crossings**.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search → goal formulation → problem formulation → execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) For each component and note, state whether it is correct or incorrect; for each incorrect one, state what is wrong.

Component / Note	Correct / Incorrect	What is wrong
State representation		
Initial state		
Actions		
Goal test		
Legality check		
Path cost		
Note N1		
Note N2		

(b) Root cause analysis: for every error, explain the underlying cause, i.e. which principle of problem formulation or of the problem-solving agent has been violated.

Error (component / note)	Root cause

(c) Give the corrected version of every faulty component or note and justify each correction.

Component / Note	Corrected formulation	Justification

(d) Apply your corrected formulation.

(i) Complete the table of boat loads available from the initial state, writing each resulting state as (M, C, boat side) and whether it is legal, with the reason.

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M		
2M		
1C		
2C		
1M + 1C		

(ii) Give one state that the student's legality check accepts although it is actually illegal.

(iii) Using your corrected path cost, state the cost of an optimal solution.

(ii)	
(iii)	

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

A student ran BFS, DFS and uniform-cost search (UCS) on the weighted undirected graph below. The start node is S and the goal node is G. **Conventions:** graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. **BFS:** do not add a node that is already explored or already in the queue. **DFS:** use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. **UCS:** write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack initially [S], top on the left)

Iteration	Node Popped	Stack after Pushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S – D – G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)

2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

(a) Examine the three traces and the remarks. List every error, giving the trace and iteration (or the remark number) and what is wrong. An error that is only carried forward counts once.

No.	Where (trace / iteration / remark)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the rule or concept that was violated and how it led to the wrong entry or conclusion.

No.	Root cause (rule or concept violated)
1	
2	
3	
4	
5	
6	
7	

(c) Write the corrected traces and justify each correction.

(i) BFS: the corrected queue after iteration 3 and the full order in which nodes are dequeued.

Queue after iteration 3	
Order of dequeued nodes	

(ii) DFS: the corrected trace.

Iteration	Node Popped	Stack after Pushing Children
1		
2		
3		
4		
5		

(iii) UCS: the corrected trace.

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1			
2			
3			
4			
5			
6			
Justification of the corrections			

(d) Apply the corrected traces.

(i) Complete the comparison table.

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS				
DFS				
UCS				

(ii) Use iteration 3 of the corrected BFS trace to explain why an explored set is needed in graph search.

(ii)	

Question 4: Search with Partial Information – Error Analysis of BeliefState Computation

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search)

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.
C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Examine the tables and the conclusions. List every error, giving the task and step (or the conclusion number) and what is wrong. An error that is only carried forward from an earlier wrong row counts once.

No.	Where (task / step / conclusion)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the concept or rule of belief-state search that was misapplied.

No.	Root cause (concept or rule misapplied)
1	
2	
3	
4	
5	
6	
7	

(c) Write the corrected tables and conclusions, with justification.

(i) Task A: the corrected belief states.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	
2	Right	
3	Suck	

(ii) Task B: the corrected belief states after each action and each percept.

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	

1	Suck		[Left, Clean]	
2	Right		[Right, Dirty]	

(iii) Corrected conclusions C1 and C2, with justification.

C1	
C2	

(d) Apply the corrected concepts.

(i) How many different belief states are possible when there are 8 physical states? Why does this make belief-state search harder than search over physical states?

(i)	

(ii) For the initial belief state {S1, S3, S5, S7}, apply the plan [Right, Suck], write the belief state after each action, and state whether it is a solution.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	
2	Suck	
	Solution? Why?	

Rubrics for Error Analysis

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	**20**	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	**30**	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	**25**	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided

Application of Concepts	**15**	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	**10**	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Question 1: Foundations of AI, Intelligent Agents and Rationality

(a) Identification of correct and incorrect statements

Statement	Correct / Incorrect	Erroneous phrase / claim

S1	Incorrect	(i) "judges a machine to be intelligent if its **internal reasoning always follows the laws of logic**" – the Turing Test judges external behaviour, not internal reasoning. (ii) "the standard example of the **'thinking rationally'** approach" – it is the standard example of the 'acting humanly' approach. (Both phrases come from one misconception, so it is counted as one error.)
S2	Incorrect	"AI winters … occurred **in the 1950s, before the 1956 Dartmouth workshop**" – wrong chronology. The follow-on claim "that workshop then **revived** the field" is carried forward from this error (the workshop founded the field; there was nothing to revive). Counted once.
S3	Incorrect	"A rational agent **always achieves the best possible outcome because it knows the actual result of every action in advance**." The conclusion "the taxi … **was not rational**" is carried forward from this claim. Counted once.
S4	Incorrect (two errors)	(i) "**Performance measure**: steering, accelerator, brake, horn" – these are actuators. (ii) "**Actuators**: safe, fast, legal and comfortable trip" – this is the performance measure. (The two PEAS components are swapped. Environment and Sensors are correct.)
S5	Incorrect (three errors)	(i) "**partially observable**" (ii) "**stochastic**" (iii) "**episodic**". The words "static" and "discrete" are correct.
S6	Correct	– (matches the definition of a model-based reflex agent)

(b) Root cause analysis

Error (statement / part)	**Root cause (misconception)**
S1	Confuses the **four approaches to AI**. The Turing Test is an operational, behavioural test (a human interrogator cannot tell the machine from a person), so it belongs to **acting humanly**. "Thinking rationally" is the laws-of-thought / logic tradition (Aristotle, syllogisms, logicist AI). The student mixed the criterion (human-like behaviour) with the internal process (logical reasoning).

S2	Confuses the **timeline / eras of AI**. The 1956 Dartmouth workshop is the birth of AI (the term was coined by McCarthy), so no "winter" can precede it. The winters followed over-hyped promises: mid-1970s (after machine-translation failure, perceptron limits and the Lighthill report) and late 1980s – early 1990s (expert-system and Lisp-machine collapse).
S3	Confuses **rationality with omniscience / perfection**. Rationality maximises **expected** performance using the percept sequence and built-in knowledge; it does not depend on the **actual** outcome, which may be affected by unforeseeable events. The student judged the agent by outcome, not by the quality of its decision.
S4-(i)	Misunderstands **PEAS**: the performance measure is the criterion of success (what the designer wants – safe, fast, legal, comfortable trip), not the physical devices the agent uses to act.
S4-(ii)	Same misunderstanding in reverse: actuators are the means by which the agent acts on the environment (steering, accelerator, brake, horn); they are not a success criterion.
S5-(i)	Mistakes **observability**. All chess pieces are visible to both players, so the complete state is available to the sensors: **fully observable**. Not knowing the opponent's future moves (or intentions) is not partial observability.
S5-(ii)	Mistakes **determinism**. A chess move has a fully determined result and there is no chance element (dice, shuffling). Opponent unpredictability makes the environment **multi-agent (competitive)**, not stochastic.
S5-(iii)	Mistakes **episodic vs sequential**. In chess the current move affects all future positions; episodes are not independent, so it is **sequential**.

(c) Corrected statements with justification

Statement	Corrected statement	Justification
S1	The Turing Test (1950) judges a machine to be intelligent if a human interrogator, after written questioning, cannot tell its answers from a human's. It is therefore the standard example of the "**acting humanly**" approach to AI.	Turing's test is purely behavioural: it requires natural-language processing, knowledge representation, automated reasoning and machine learning to produce human-like behaviour, but it says nothing about whether the internal reasoning is logical.

S2	The **1956 Dartmouth workshop** coined the term “artificial intelligence” and launched AI as a field. The “AI winters” (periods of reduced funding and interest) came **afterwards**: the first in the mid-1970s and the second from the late 1980s to the early 1990s.	The workshop is the starting point of AI; winters are reactions to unmet expectations and can only occur after the field exists. Only the “coined the term” part of the original statement was correct.
S3	A rational agent selects the action that is **expected** to maximise its performance measure, given its percept sequence and built-in knowledge; it does not need to know actual outcomes in advance. Rationality is not omniscience, so a taxi whose route was later blocked by an **unforeseeable** landslide was still rational.	Rationality is judged by the decision given the available information, not by luck. Omniscience (knowing actual outcomes) is impossible in practice; the landslide was not in the percept sequence.
S4	**Performance measure:** safe, fast, legal, comfortable trip. **Environment:** roads, other traffic, pedestrians, weather. **Actuators:** steering, accelerator, brake, horn. **Sensors:** cameras, GPS, speedometer.	P = how success is measured, E = where the agent operates, A = what it acts with, S = what it perceives with. The student only swapped P and A; E and S were already correct.
S5	Chess played without a clock is a **fully observable, deterministic, sequential, static and discrete** task environment (also multi-agent and competitive).	All pieces are visible; moves have fixed outcomes; each move affects later positions; nothing changes while the agent deliberates (no clock); finite set of states and moves.

S6	**No change.**	A model-based reflex agent maintains internal state, updated with (1) knowledge of how the world evolves independently of the agent and (2) how its own actions affect the world. This lets it handle partial observability, which is exactly what the statement claims.

(d) Application to the self-driving taxi

(i) Was the taxi's choice rational? Yes. Using the four factors that decide rationality:

Factor	Application to the taxi	Verdict
1. Performance measure	Success = safe, fast, legal, comfortable trip. A highway route reported clear gives the best expected speed and comfort.	Satisfied
2. Agent's prior knowledge of the environment	Knows the road map, traffic rules and typical highway behaviour. Landslides on this route were not predictable from this knowledge.	Satisfied
3. Actions the agent can perform	Could choose between routes, speeds, lanes, stop or re-route. The highway was the best-expected option among them.	Satisfied
4. Percept sequence to date	Every available traffic report showed the highway clear. The landslide is not in the percept sequence.	Satisfied

Since the action maximised **expected** performance given what the taxi knew and perceived, it was rational. The bad outcome was bad luck, not bad reasoning. (It would have been irrational only if the taxi had ignored available reports, i.e. skipped information gathering.)

(ii) Properties of the task environment that make perfect foresight impossible: the environment is **partially observable** (the taxi cannot sense the state of the whole road network or hillsides ahead), **stochastic** (outcomes such as landslides, accidents and traffic are uncertain), and **dynamic** (the world changes while the agent deliberates). It is also sequential, continuous and multi-agent, which adds to the uncertainty.

(iii) Simple reflex vs model-based reflex taxi. A simple reflex taxi decides only from the **current** percept (condition–action rules). Example: it wants to change lanes, but a car is in its blind spot; the

current camera frame shows nothing, so the reflex rule “lane clear → change lane” fires and it may cause a collision. (Similarly it cannot tell from one frame whether the car ahead is just showing tail lights or is actually braking.) A model-based reflex taxi keeps an **internal state** – the last observed positions and speeds of nearby vehicles, its own recent actions (e.g. it has been accelerating, it signalled a turn) – updated with a model of **how the world evolves** (a car in the blind spot keeps moving at similar speed) and **how its own actions affect it**. It therefore knows a car is still beside it and waits.

Question 2: Problem-Solving Agents and Problem Formulation

(a) Component-wise check

Component / Note	Correct / Incorrect	What is wrong
State representation	Incorrect	(M, C) omits the **boat position**. The same (M, C) can occur with the boat on either bank, and the legal moves differ. State must be (M, C, B).
Initial state	Incorrect	"(3, 3)" does not say where the boat is. It should be (3, 3, L). (Carried forward from the state-representation error; counted once.)
Actions	Correct	The five loads 1M, 2M, 1C, 2C, 1M+1C are all the possible non-empty loads of at most two people. (Each load is applicable only if those people are on the boat's current bank.)
Goal test	Incorrect	"M = C" is true in many non-goal states, e.g. (3, 3), (2, 2), (1, 1). The goal is everyone on the right bank.
Legality check	Incorrect	It checks only the **left bank**. The right bank must also have no missionaries or at least as many missionaries as cannibals.
Path cost	Incorrect	"Number of people carried" does not match the objective, which is the **fewest crossings**. Each crossing should cost 1.
Note N1	Incorrect	Wrong order of the problem-solving agent's steps: it should be goal formulation → problem formulation → search → execution.
Note N2	Incorrect	The figures 16 / 10 ignore the boat. With the boat there are $4 \times 4 \times 2 = 32$ states, of which **20** are legal (10 legal (M, C) pairs × 2 boat sides). (Arises from the same omission as the state representation, but the numbers must be corrected.)

(b) Root cause analysis

Error (component / note)	Root cause

State representation (and Initial state, N2 carried)	Violates the principle that a **state must contain all information needed to determine the legal actions and their results**. The boat's position decides which bank people can leave from, so leaving it out gives an incomplete abstraction, an ambiguous initial state and a wrong state-space size.
Goal test	A goal test must be a **precise test of the goal condition** (all six people on the far bank). "M = C" is a safety (balance) property, not the objective; the student confused a constraint with the goal.
Legality check	A constraint ("missionaries never outnumbered **on either bank**") was applied to only one bank. A state-validity test must enforce the condition on the whole state, so the right bank was ignored.
Path cost	Path cost must **encode the performance measure/objective**. The objective is the fewest crossings, so cost should be a step cost of 1 per action; counting people rewards a different objective and would not give the optimal solution.
N1	Misunderstands the **problem-solving agent's cycle**: goal formulation limits the objectives, problem formulation decides which states and actions to consider to reach that goal, and only then can search be run and the solution executed. Search cannot start without a formulated problem.
N2	Counting the state space from an incomplete state representation; legal-state counting must also include the boat's side, and legality must be checked on both banks.

(c) Corrected formulation

Component / Note	**Corrected formulation**	**Justification**
State representation	(M, C, B): M, C = missionaries and cannibals on the **left** bank; B ∈ {L, R} = side of the boat.	Contains all the information needed to find the legal actions; the right-bank counts follow as (3 − M, 3 − C).
Initial state	(3, 3, L)	Everyone and the boat start on the left bank.

Actions	No change: carry 1M, 2M, 1C, 2C or 1M+1C from the boat's bank to the other bank (only if available there).	The set of loads is already correct and complete.
Goal test	State = (0, 0, R)	All six people on the right bank and the boat with them.
Legality check	State legal iff (M = 0 or M ≥ C) **and** (M = 3 or M ≤ C). (Right bank: 3 − M = 0 or 3 − M ≥ 3 − C ⇔ M = 3 or C ≥ M.)	Enforces "missionaries never outnumbered" on both banks; a bank with no missionaries is safe.
Path cost	Step cost = **1 per crossing**; path cost = number of crossings.	Matches the objective "fewest crossings".
Note N1	Goal formulation → problem formulation → search → execution.	The goal must be set first, then the problem (states and actions) is formulated; search produces the action sequence; execution carries it out.
Note N2	32 states in total; 20 legal (not all reachable from the initial state).	State = (M, C, B) with 4 × 4 × 2 = 32 combinations. Legal pairs (M, C): (0,0), (0,1), (0,2), (0,3), (3,0), (3,1), (3,2), (3,3), (1,1), (2,2) = 10; ×2 boat sides = 20.

(d) Application of the corrected formulation

(i) Boat loads from the initial state (3, 3, L):

Boat load	**Resulting state (M, C, boat)**	**Legal?**	**Reason**
1M	(2, 3, R)	No	Left bank: 2 missionaries < 3 cannibals.
2M	(1, 3, R)	No	Left bank: 1 missionary < 3 cannibals.
1C	(3, 2, R)	Yes	Left 3M, 2C (safe). Right 0M, 1C (no missionaries, safe).
2C	(3, 1, R)	Yes	Left 3M, 1C (safe). Right 0M, 2C (safe).
1M + 1C	(2, 2, R)	Yes	Left 2M, 2C (equal). Right 1M, 1C (equal).

(ii) A state accepted by the student's check but actually illegal: (M, C, B) = (2, 1, ·), e.g. (2, 1, R). The left bank is fine (C = 1 ≤ M = 2), but the right bank has 1 missionary and 2 cannibals, so the missionary is outnumbered. (Another example: (1, 0, ·), where the right bank has 2M and 3C.)

(iii) Cost of an optimal solution = 11 crossings (step cost 1). One optimal sequence:

Step	Crossing	State after (M, C, B)	Bank check
1	1M + 1C →	(2, 2, R)	Left 2M2C, right 1M1C
2	1M ←	(3, 2, L)	Right 0M1C safe
3	2C →	(3, 0, R)	Right 0M3C safe
4	1C ←	(3, 1, L)	Right 0M2C safe
5	2M →	(1, 1, R)	Left 1M1C, right 2M2C
6	1M + 1C ←	(2, 2, L)	Left 2M2C, right 1M1C
7	2M →	(0, 2, R)	Left 0M, right 3M1C
8	1C ←	(0, 3, L)	Right 3M0C
9	2C →	(0, 1, R)	Right 3M2C
10	1C ←	(0, 2, L)	Right 3M1C
11	2C →	(0, 0, R)	Goal

Note: 11 crossings is the known minimum for 3 missionaries, 3 cannibals and a boat for 2. With the student's "people carried" cost the same plan would be priced at 18, which does not measure the stated objective.

Question 3: Searching for Solutions and Uninformed Search

Adjacency used (alphabetical, with edge costs):

Node	Neighbours
S	C(3), D(3)
A	B(2), C(6)
B	A(2), D(4), G(3)
C	A(6), S(3)
D	B(4), E(2), G(6), S(3)
E	D(2), G(3)
G	B(3), D(6), E(3)

(a) List of errors

No.	Where (trace / iteration / remark)	What is wrong
1	BFS, iteration 3	Queue after expanding D is written [B, E, G, **A**]. BFS is FIFO: new nodes join the **back**, so A (already waiting) must stay in front: [A, B, E, G].
2†	BFS, iterations 4–6 and result line	Carried forward from error 1: B is dequeued before A, A is never dequeued, and the result says 6 nodes removed. Correct dequeue order is S, C, D, A, B, E, G = **7** nodes. (Counted once with error 1; the path S–D–G and cost 9 are correct.)
3	DFS, iteration 1 (repeated in iteration 2)	Children pushed in the wrong order: stack [D, C] makes D pop first. Alphabetically first child C must be on top: [C, D]. Because of this the whole trace and its result (S–D–G, cost 9, 3 nodes) are wrong; those are carried forward.
4	UCS, iteration 4	Expanding E (g = 5) gives G with g = 5 + 3 = 8, which is cheaper than G:9(D). The entry must be **replaced** by G:8(E); the student left G:9(D). Iterations 5–7 and the result (cost 9, 7 nodes) are carried forward.

5	Remark R1	False: in graph search BFS needs the explored set (or an equivalent check); the graph is undirected and cyclic, so nodes are reachable again by other routes.
6	Remark R2	False: the strategies do not agree once traces are corrected (BFS cost 9, DFS cost 14, UCS cost 8). Agreement of final answers would also not prove that traces are correct, and only UCS is optimal.
7	Remark R3	False: the number of nodes removed is not the criterion of "best". Corrected, DFS removes 5 nodes but returns the worst path (cost 14); DFS is not optimal. (The count of 3 was also wrong.)

† Row 2 is a consequence of row 1, so it is not an independent error.

(b) Root cause analysis

No.	Root cause (rule or concept violated)
1	Violation of the **FIFO queue discipline** of BFS (nodes are expanded level by level). The student inserted new children ahead of an already queued node, giving a wrong dequeue order (B before A).
2	Error propagation from row 1: wrong queue content → wrong node dequeued at iteration 4 and a wrong node count.
3	Violation of the stated **stack convention**: with top on the left and the alphabetically first child to be popped first, the first child must be placed on top. Pushing in reverse order changed the exploration order from C-first to D-first, so DFS jumped to the deepest-wrong branch and reported a wrong path.
4	Failure to apply the UCS **"replace if cheaper path to a frontier node"** rule (relaxation). UCS orders nodes by path cost g(n), and the frontier must always hold the cheapest known path to each node; ignoring the cheaper route to G via E gave a non-optimal cost 9.
5	Misunderstanding of **tree search vs graph search**. "BFS never revisits a node" holds only because of the explored set / frontier check. Without it, nodes are re-added through cycles and alternative paths (e.g. S from C and D).
6	Confusion about **optimality**: BFS is optimal only for equal step costs (it minimises edges, not cost); DFS has no optimality guarantee; only UCS is optimal for non-uniform costs. Equal final outputs were taken as proof, and the earlier errors (3 and 4) are what produced the false agreement.

7	Mixing up a **cost/efficiency measure with solution quality**. Fewest removals reflects time/space effort, not optimality; a strategy cannot be judged best from node count alone.

(c) Corrected traces

(i) BFS (queue initially [S])

Queue after iteration 3	[A, B, E, G]
Order of dequeued nodes	S, C, D, A, B, E, G

Full corrected BFS trace for reference:

Iteration	Node dequeued	Queue after expansion	Explored set / note
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}; S explored, A added
3	D	[A, B, E, G]	{S, C, D}; B, E, G join the back; S explored, so not added
4	A	[B, E, G]	{S, C, D, A}; B already in queue, C explored
5	B	[E, G]	{S, C, D, A, B}; D explored, G already in queue
6	E	[G]	{S, C, D, A, B, E}
7	G (goal)	[ ]	Goal test on removal. Path S–D–G (G was added from D), cost 3 + 6 = 9

(ii) DFS (stack top on the left)

Iteration	Node popped	Stack after pushing children	Children pushed / reason
1	S	[C, D]	C on top so it is popped first
2	C	[A, D]	S explored; A pushed on top
3	A	[B, D]	C explored; B pushed on top
4	B	[G, D]	A explored; D already on the stack, not pushed; G pushed on top

5	G (goal)	[D]	Goal test on pop. Path S–C–A–B–G, cost 3 + 6 + 2 + 3 = 14

(iii) UCS (entries as Node:g(parent))

Iteration	Node expanded	g(n)	Priority queue after expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), **G:8(E)**, A:9(C) (G:9(D) replaced by the cheaper G:8(E))
5	B	7	G:8(E), A:9(C) (A via B = 9, not cheaper than 9, so no change; G via B = 10 is not cheaper)
6	G (goal)	8	– Path S–D–E–G, cost 3 + 2 + 3 = 8

Justification of the corrections.

- **BFS:** FIFO means D's children join behind A, giving [A, B, E, G]. Then A (not B) is dequeued fourth. The path S–D–G (cost 9) was right, but 7 nodes (not 6) leave the frontier before G is removed.
- **DFS:** with the correct push order C is explored first. Following the explored-set and "already on stack" rules, the search goes S → C → A → B → G, so the result is S–C–A–B–G, cost 14, with 5 nodes removed.
- **UCS:** the cheaper route to G through E (8 < 9) replaces the entry, and G is removed with g = 8 before A is expanded; 6 nodes are removed and the optimal cost is 8.

(d) Application of the corrected traces

(i) Comparison table

Strategy	Path found	Path cost	Nodes removed from frontier	Optimal here?
BFS	S–D–G	9	7	No (optimal cost is 8)
DFS	S–C–A–B–G	14	5	No
UCS	S–D–E–G	8	6	Yes

(ii) Why an explored set is needed (using BFS iteration 3). When D is expanded its neighbours are B, E, G and **S**. S has already been dequeued and expanded (it is in the explored set {S, C, D}), so it is not added. Without the explored set, S would be added again: [A, B, E, G, S]. Expanding it again would re-add C and D, and the search would loop forever on this cyclic undirected graph, wasting time and memory (and in other cases failing to terminate). The same check also stops B and G being queued repeatedly through A, B and E later on.

Question 4: Search with Partial Information – Belief States

Physical states: S1 (L,D,D), S2 (R,D,D), S3 (L,D,C), S4 (R,D,C), S5 (L,C,D), S6 (R,C,D), S7 (L,C,C), S8 (R,C,C). Goal states: **S7 and S8** (both squares clean). Predicted belief state = apply the action to every state in the belief state; updated belief state = keep only the states consistent with the percept.

(a) List of errors

No.	Where (task / step / conclusion)	What is wrong
1	Task A, step 3 (Suck)	Belief {S6, S8} is wrong. Suck in the Right square: S6 (R,C,D) → S8 (R,C,C); S8 stays S8. Correct belief state: **{S8}**. (The student did not apply the action to S6.)
2	Conclusion C1	False. The final belief state is the singleton {S8}, a goal; the plan [Suck, Right, Suck] is a solution. Also the general claim "sensorless problems cannot be solved" is false; they can be solved by **coercion** (actions shrink the belief state). (Even the claim "never knows where it is" is wrong – both states were in the Right square.)
3	Task B, step 0, percept update	{S1, S2, S3, S6} is wrong. Percept [Left, Dirty] keeps only states with the agent in Left **and** left square dirty: **{S1, S3}**. S2 and S6 have the agent in the Right square.
4†	Task B, step 1, predicted state	{S4, S5, S7, S8} is only what follows from the wrong belief {S1, S2, S3, S6} (S2 → S4, S6 → S8). Carried forward from error 3; counted once. Correct: {S5, S7}.
5	Task B, step 2, predicted state	{S6, S7} is wrong. After Right, S5 (L,C,D) → S6 (R,C,D) correctly, but S7 (L,C,C) → **S8** (R,C,C), not S7 (the agent moves to the Right square). Correct predicted belief state: **{S6, S8}**.
6	Conclusion C2	The belief {S6} is a singleton (knowing location and status is true), but "the goal has been reached" is false: S6 = (R, C, D), the right square is still **dirty**. One more Suck is needed to reach {S8}.
7	Conclusion C3	False. A belief state is a **set** of physical states, so there are $2^8 = 256$ possible belief states (255 non-empty), not 8.

† Row 4 is a consequence of row 3, so it is not an independent error. (Task A steps 1 and 2 and Task B step 1 percept and step 2 percept updates are correct given their inputs.)

(b) Root cause analysis

No.	Root cause (concept or rule misapplied)
1	The **prediction (transition) step** was not applied to every physical state in the belief state. S6 must be mapped through Suck in the right square to S8; the student left it unchanged, as if the action had no effect.
2	Misunderstanding of **solutions in belief-state space**: a plan is a solution if the final belief state is a subset of the goal states, whatever the starting uncertainty. Sensorless planning works by coercing the world into a known state; the student assumed that uncertainty at start (or a non-singleton) is permanent.
3	Misapplied the **percept (update) step**: the updated belief state is the predicted belief state **filtered** by the percept. The percept [Left, Dirty] contains the location and so excludes every state with the agent on the Right; the student kept states that contradict the percept (S2, S6).
4	Error propagation: the prediction step was applied correctly to a wrongly updated belief state.
5	Incorrect **transition model**: moving Right changes the agent's location in every physical state, including S7 (L,C,C) → S8 (R,C,C). The student applied Right to S5 but not to S7.
6	Confusion of **knowledge with goal achievement**: the **goal test on a belief state** requires every state in it to be a goal state. A singleton belief state tells the agent exactly where it is, but S6 is not a goal state.
7	Mistaking a belief state for a physical state. Belief states are **subsets (the power set)** of physical states, 2^N of them; searching them is not equal in size to searching N physical states.

(c) Corrected tables and conclusions

(i) Task A

Step	Action	Belief state after action
0 (start)	–	{S1, S3, S5, S7}
1	Suck	{S5, S7} (S1 → S5, S3 → S7, S5 and S7 unchanged)
2	Right	{S6, S8} (S5 → S6, S7 → S8)
3	Suck	**{S8}** (S6 → S8, S8 unchanged)

(ii) Task B

Step	Action	After action (predicted)	Percept	After percept (updated)
0 (start)	–	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	**{S1, S3}** (agent Left and left square dirty)
1	Suck	{S5, S7} (S1 → S5, S3 → S7)	[Left, Clean]	{S5, S7} (both have agent Left and left clean)
2	Right	**{S6, S8}** (S5 → S6, S7 → S8)	[Right, Dirty]	{S6} (S8 has the right square clean, so it is removed)

(iii) Corrected conclusions

	Corrected conclusion	Justification
C1	Task A: the final belief state is {S8}, a singleton goal state, so the plan [Suck, Right, Suck] **is a solution**. Sensorless problems **can** be solved.	All states in the final belief state are goal states. The plan works from every starting state S1, S3, S5, S7: Suck cleans the left square, Right moves to the right square, Suck cleans the right square. This is coercion – no sensing is needed.
C2	Task B: the belief state {S6} is a singleton, so the agent knows its location (Right) and dirt status (left clean, right dirty), but the goal has **not yet been reached**. One more Suck gives {S8}.	The goal test on a belief state requires every member to be a goal state (S7 or S8). S6 = (R,C,D) is not a goal state because the right square is dirty.

(d) Application of the corrected concepts

(i) Number of belief states. A belief state is any subset of the 8 physical states, so there are 2 = **256** belief states (255 if the empty set is excluded). Searching them is harder than searching the 8 physical states because the belief-state space grows **exponentially** (2^N for N physical states) instead of linearly, and each belief state is a set of up to N states that must be stored, compared and transformed. Even a modest increase in physical states makes the belief-state space huge, whereas a physical-state space would stay small.

(ii) Plan [Right, Suck] from {S1, S3, S5, S7}:

Step	Action	Belief state after action
0 (start)	–	{S1, S3, S5, S7}

1	Right	{S2, S4, S6, S8} (S1 → S2, S3 → S4, S5 → S6, S7 → S8)
2	Suck	{S4, S8} (S2 → S4, S4 → S4 since the right square is already clean, S6 → S8, S8 → S8)

Solution? No. The final belief state {S4, S8} contains S4 = (R, D, C), which is not a goal state because the left square is still dirty. A plan is a solution only if every state in the final belief state is a goal state. This plan never cleans the left square; the correct plan must clean it first, e.g. [Suck, Right, Suck].